

CROWN AND BRIDGE

Lecture five

Full-metal crown preparation

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Full metal crown :

It is one of the most commonly used types of crowns.

It can be used as a single unit or as a retainer for a bridge.

It has better retention and resistance to displacement than other partial coverage crowns as 3/4 crown because all the axial and occlusal walls are included.



Full metal crown :

Due to metal strength it may be used in simple crown or as a retainer for long-span bridges.

Because it made of metal, it should be used when the patient doesn't mind the appearance of metal or when esthetic is not a factor.

Used as a retainer for F.P.D, especially when we have a small abutment tooth with long span edentulous area to overcome the occlusal forces and prevent bridge displacement.

Types of metal alloys used for full metal crown

1. High noble alloys (gold alloys).
2. Low noble alloys (silver-palladium and gold-palladium alloys).
3. Non-noble alloys (Nickel-chromium alloy).



Advantages of full metal crown

1. Greater retention and strength.
2. High resistance to deformation.
3. Modification of axial tooth contour is possible
4. More conservative than other types of full crown
such as porcelain fused to metal and all ceramic
crowns.

Disadvantages of full metal crown

1. Extensive tooth structure removal as compared with partial crown such as 3/4crown.
2. Difficulty to test the vitality of the tooth especially by electrical pulp tester.
3. May interfere with taste.
4. Display of metal.

Indications for full metal crowns

1. Teeth with extensive caries or large amalgam restorations in order to protect the remaining tooth structure and amalgam from fracture (posterior teeth protection).
- 2-Teeth where maximum retention and resistance is needed as in case of short crowns (little occlusal reduction) .



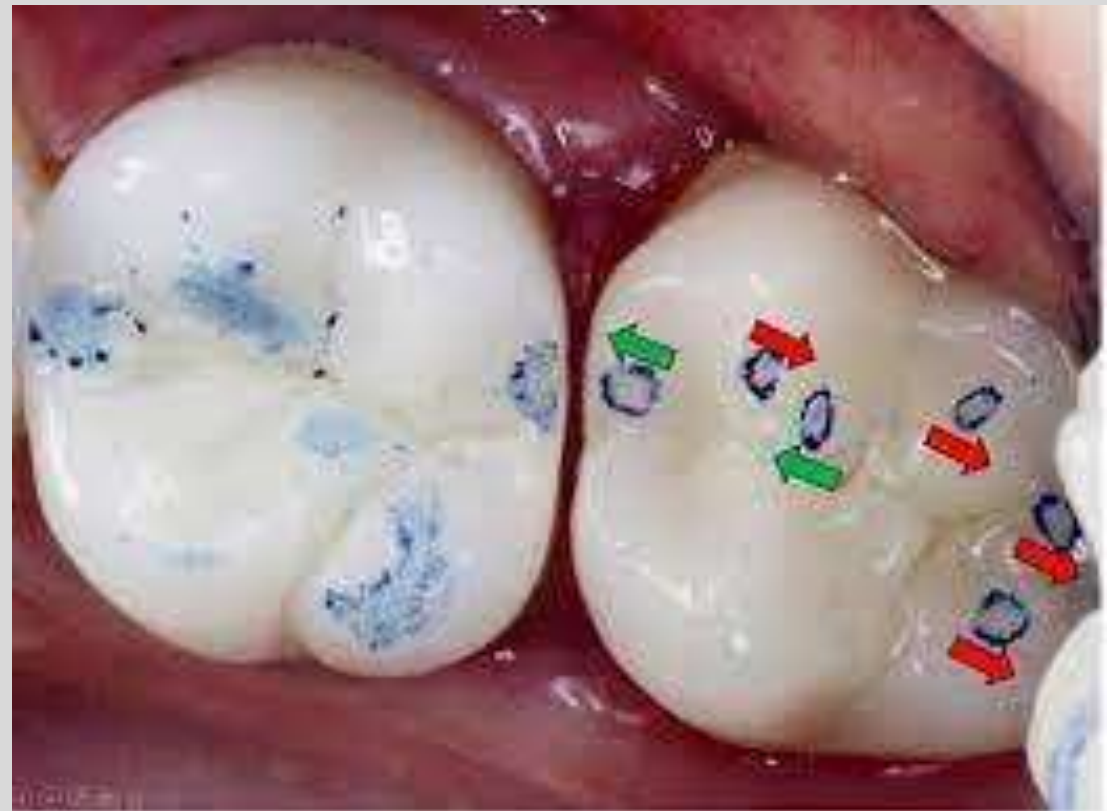
Indications for full metal crowns

- 3-As a bridge retainer for metal-ceramic bridges.
- 4-Recontouring of posterior teeth for occlusion purposes.
- 5- Recontouring of teeth receiving clasps for removable partial denture.



Indications for full metal crowns

- 6- Endodontically treated teeth.
- 7- Correction of minor inclination or big interdental space.



Contra-Indications

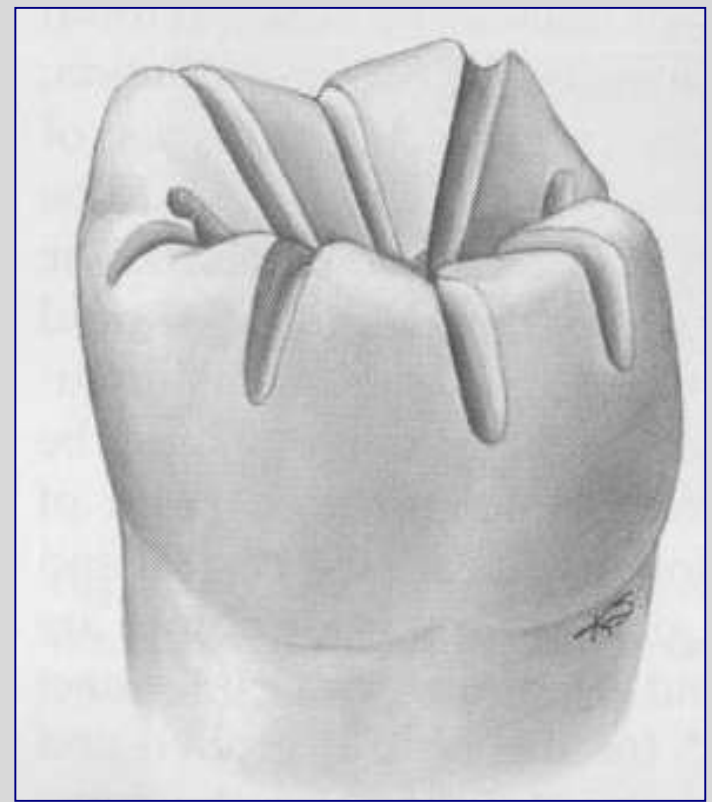
- 1-If high esthetic is needed.
- 2-If conservative crown types could be used like 3/4 crown where buccal surface left untouched .
- 3- In case of uncontrolled caries.

Steps for preparation

Depth Orientation groove:

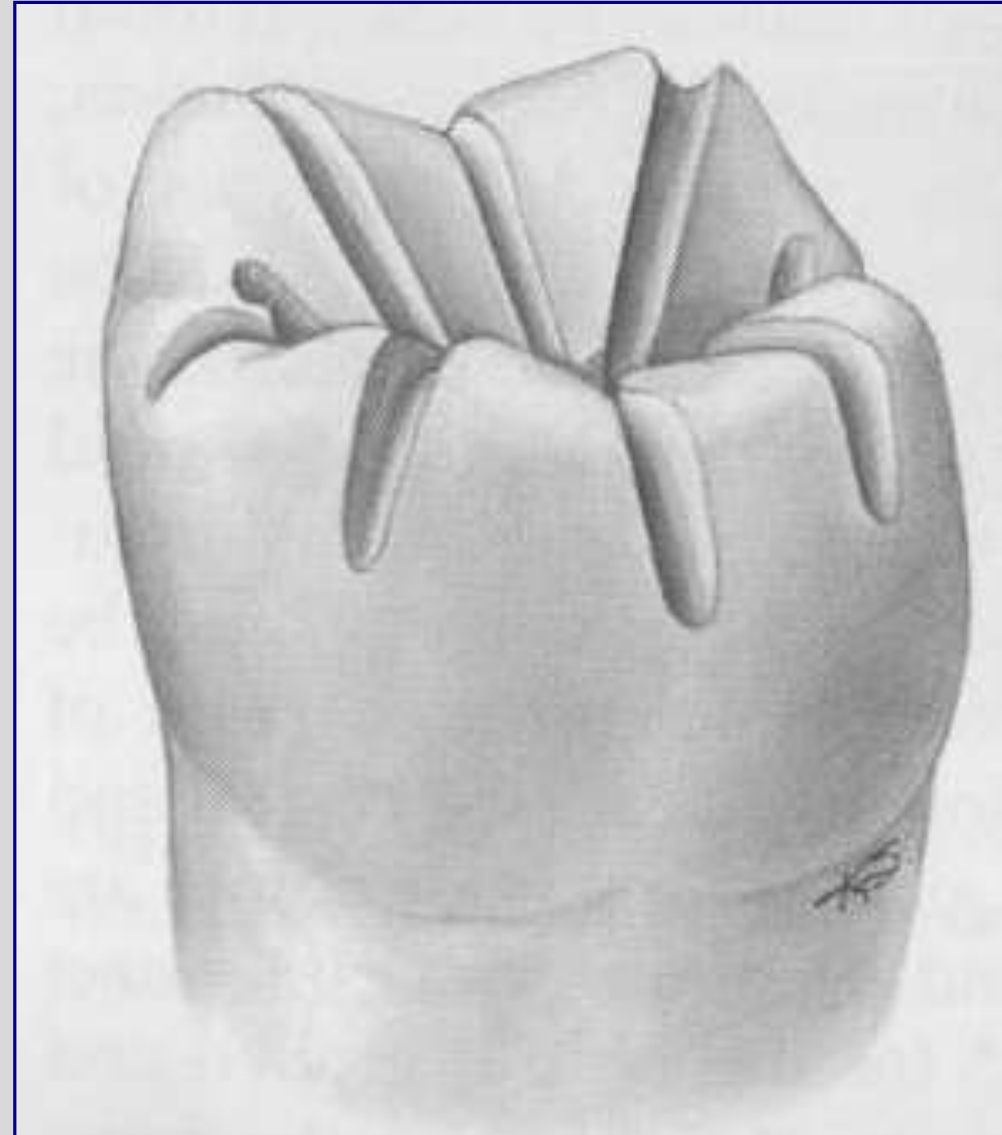
This is a groove that is placed on a surface of an **unprepared** tooth to act as a guide or reference to determine the required amount of tooth structure to be removed.

If preparation is done without these grooves under or over preparation may happened.



Occlusal surface preparation

- 1- Depth orientation grooves made on the occlusal surface by a fissure bur following the inclines of the cusps.
- 2- Cut the area between the grooves starting with any half of the occlusal surface and then prepare the other half.



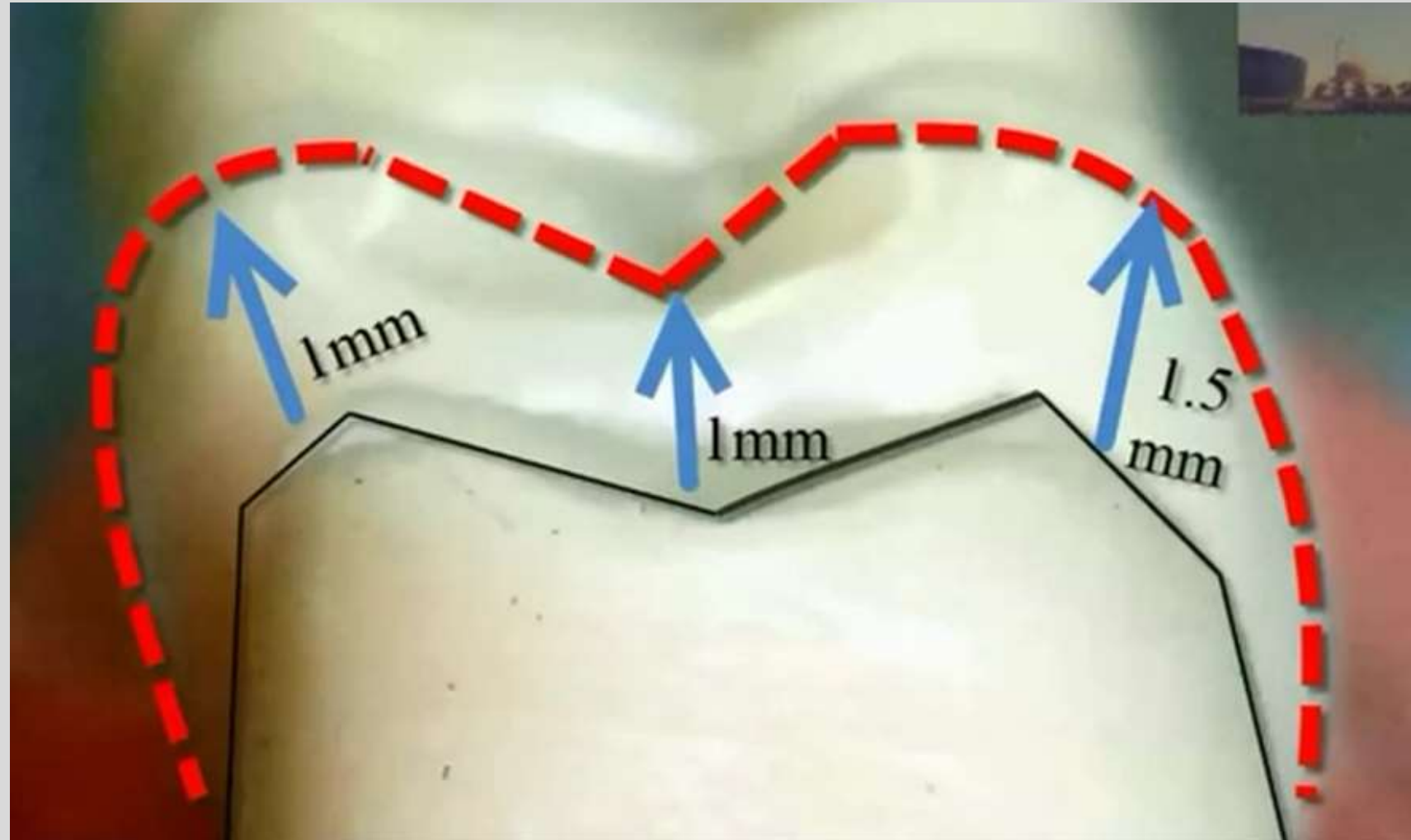
The sequence of the occlusal surface preparation is as follows:

1. Depth orientation grooves (D.O.G) are prepared in the occlusal surface by a fissure bur to follow the inclines of the cusps. A D.O.G is prepared in each cusp extending from the cusp tip to the central groove, which represents the deepest part of the occlusal surface. The depth of each groove corresponds to the diameter of the fissure bur used. i.e. a fissure bur with 1.5 mm diameter is used to prepare D.O.G on the functional cusps, while a fissure bur with 1 mm diameter is used to prepare D.O.G on the non-functional cusps.

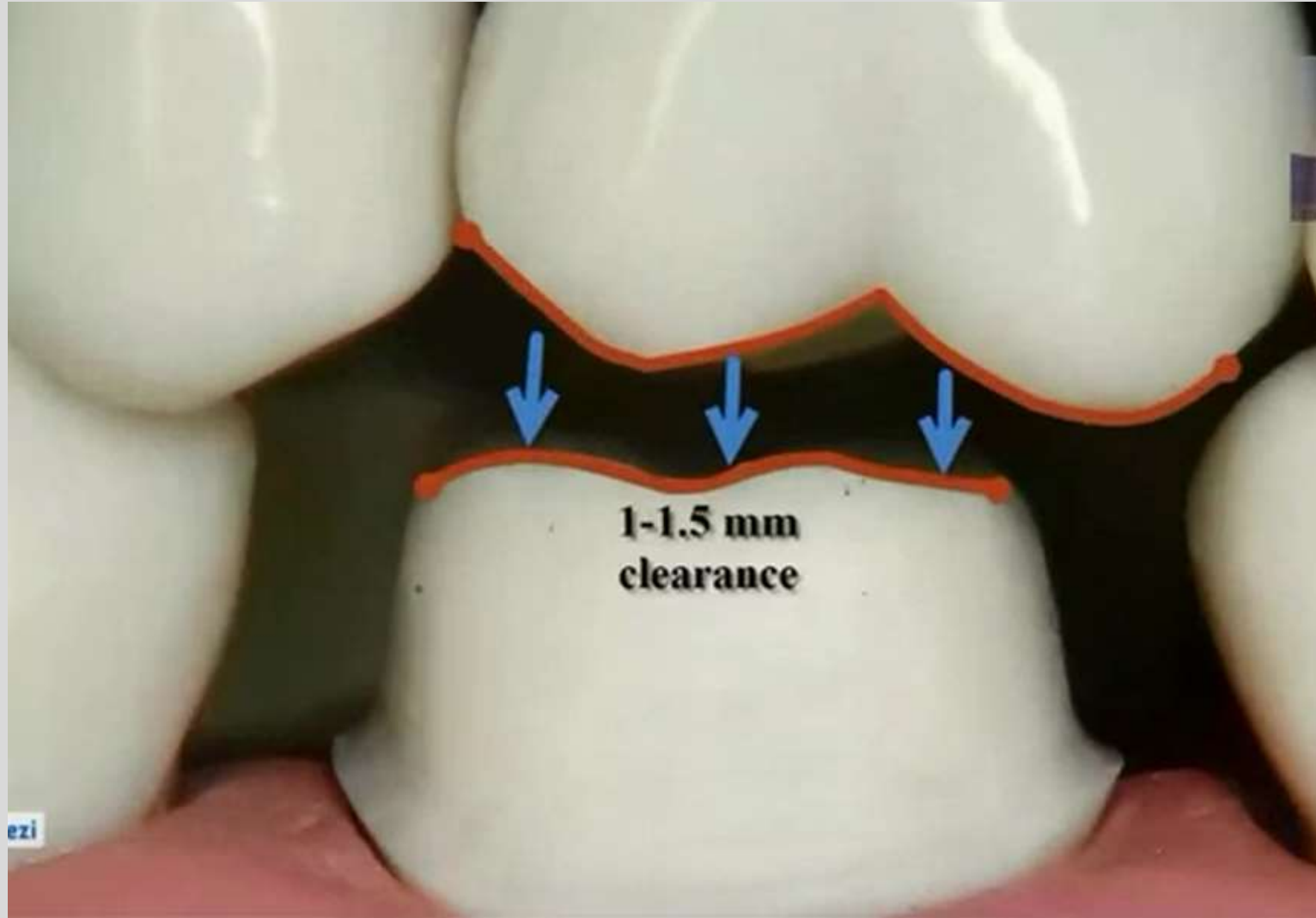
2. Any tooth structure between D.O.G should be removed following the normal contour of the cusps.
3. A wide bevel is placed on the functional cusps.
4. The occlusal clearance is then checked in centric & eccentric occlusal relations.



5- The occlusal preparation in the functional cusps should be 1.5mm (contra bevel to the buccal), and at least 1mm for the nonfunctional cusps.



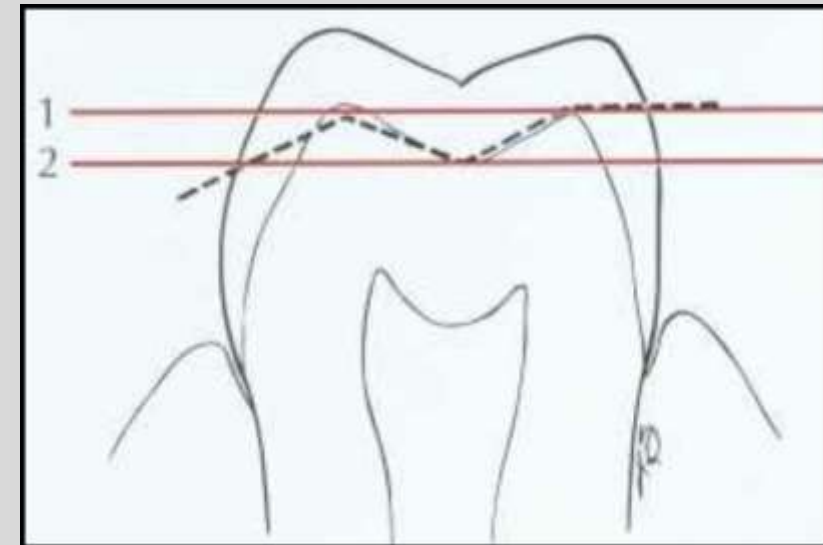
6- The occlusal clearance (1-1.5) should be kept in all occlusal movements (centric, lateral and protrusive movements).



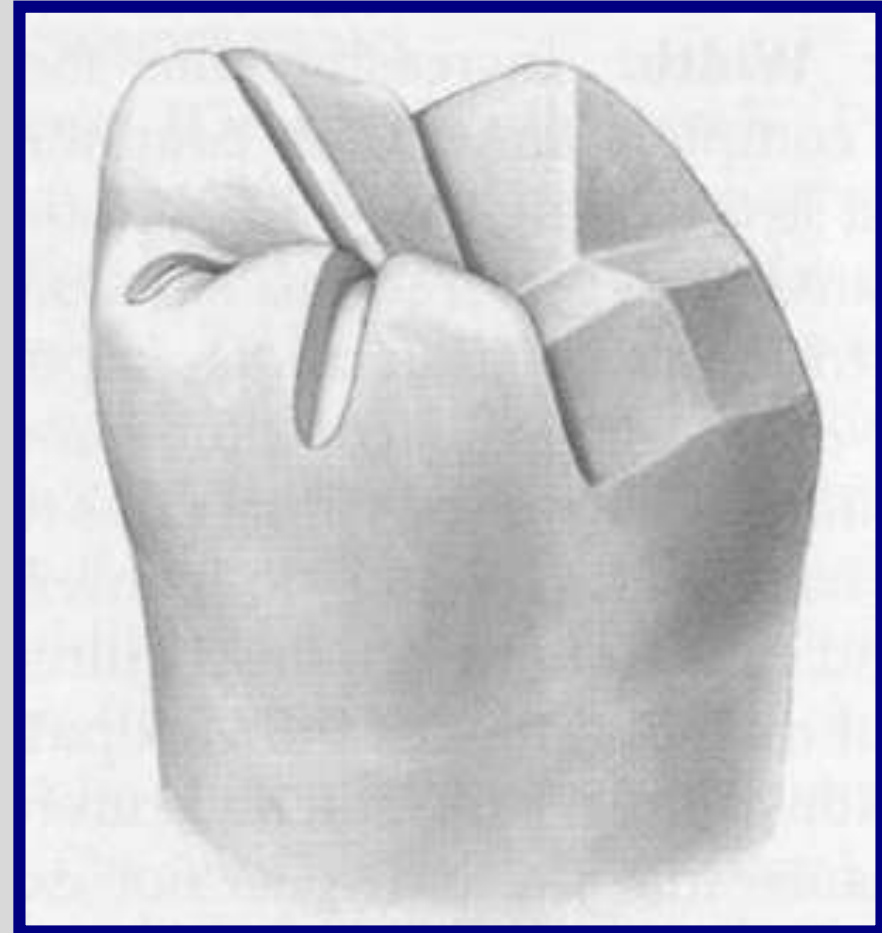
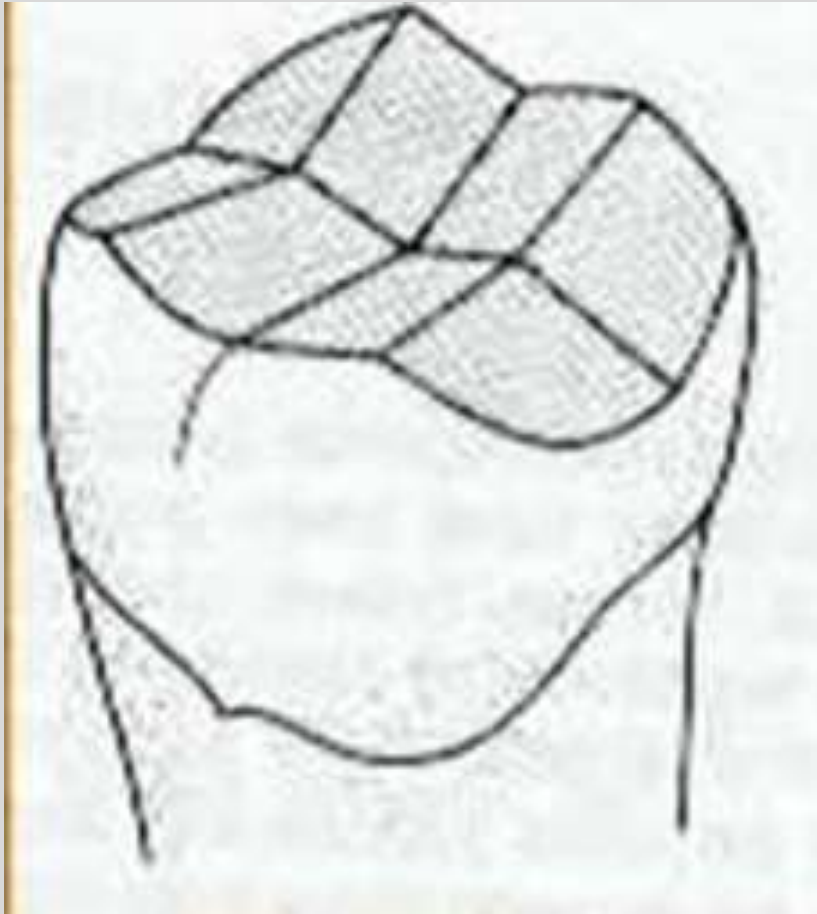
Planar occlusal reduction (anatomical reduction):

following the geometric inclined planes of the occlusal surface should be done for the following objectives:

- To provide a restoration with uniform thickness.
- To preserve the tooth structure (axial wall length).



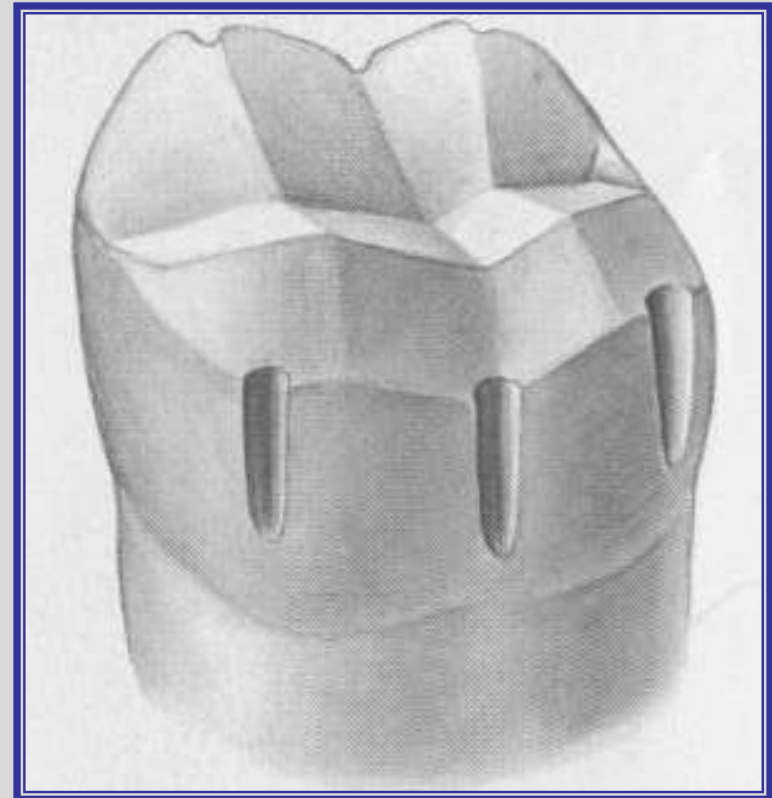
-To improve the retention- resistance features of the preparation.



Buccal and lingual preparations:

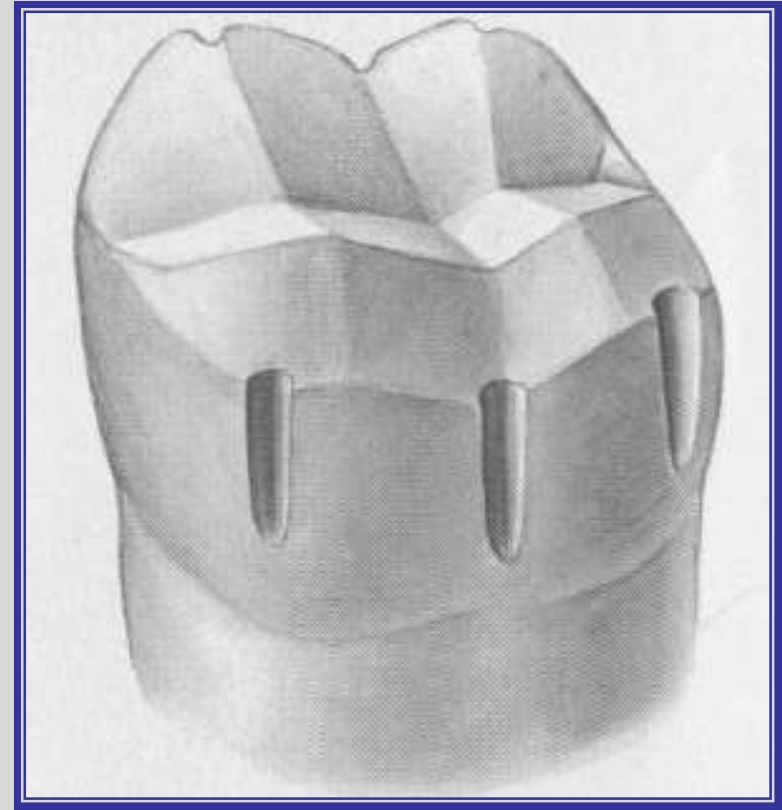
1. Three D.O.G with 1 mm depth are prepared in the buccal surface of the tooth, one placed in the center of the wall and one in each medial and distal transitional line angles. These grooves are prepared parallel to the long axis of the tooth or to the proposed path of insertion of the restoration.

Depth grooves applied by a narrow round-end fissure bur so that all the bur is embedded in the tooth structure and the long axis of the bur must be parallel with long axis of the tooth.



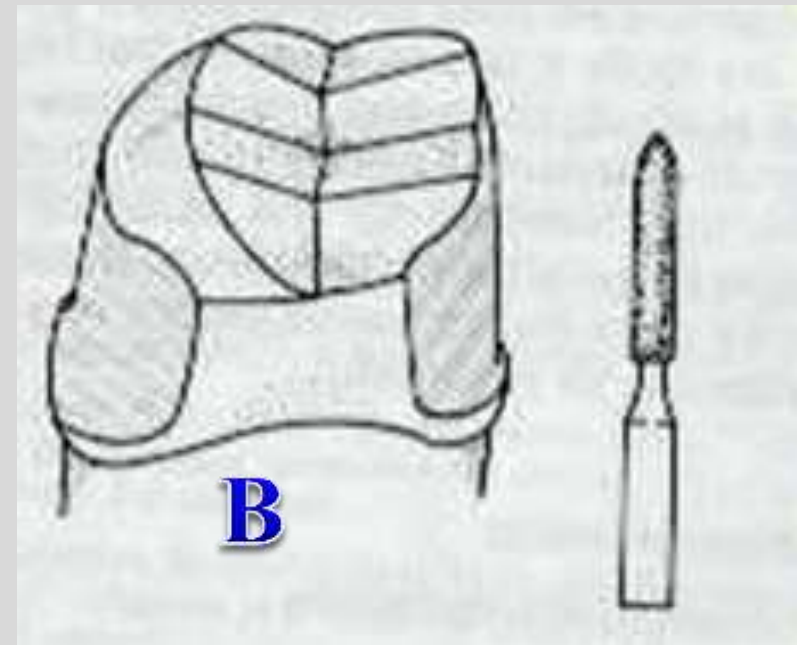
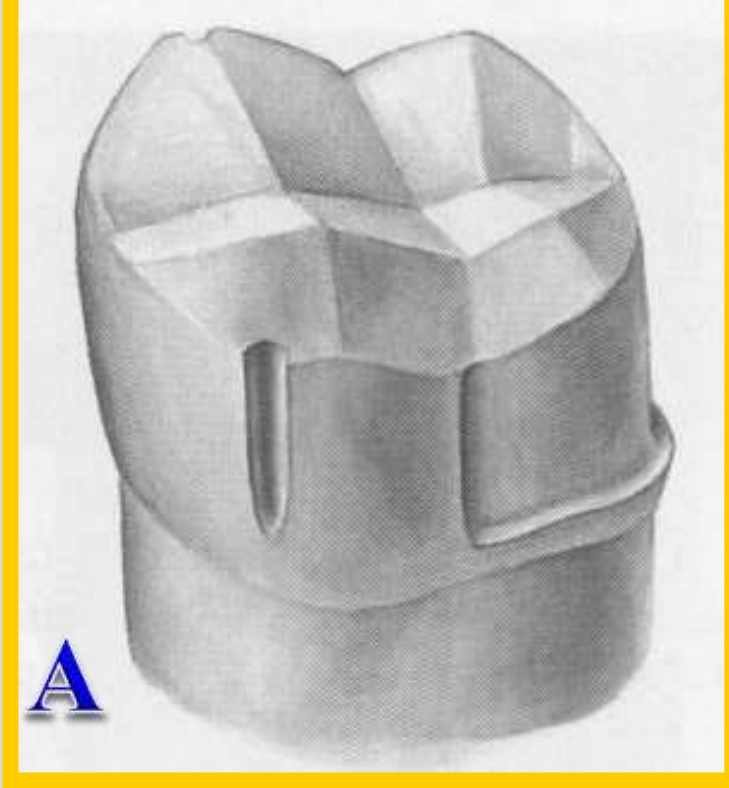
Buccal and lingual preparations:

2. Move the bur mesially and distally following the curvature of this surface to remove any islands of tooth structure between D.O.G. The gingival extent of the preparation will determine the position of the margin (whether to be placed supragingival, which is preferable, or there is a need to extend the finishing line subgingivally. A round-end tapered fissure bur is used during axial reduction to obtain continuous, smooth chamfer finishing line.



NOTE: in figure (A) the occlusal third of buccal surface was prepared with the occlusal preparation (called functional cusp contra-bevel),

while in figure (B) the buccal surface was prepared in two- step reduction technique were the occlusal third of the buccal surface will be prepared as second step after finishing the first step.



Lingual surface preparation

The preparation of the lingual surface is the same as that of the buccal surface (except it is prepared in one step reduction).

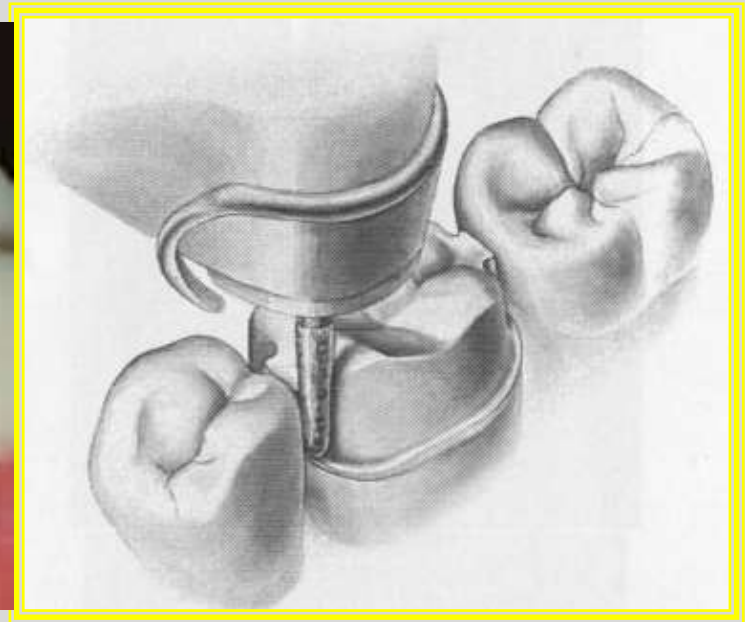
NOTE: buccal and lingual or (palatal) surfaces may be prepared in one or two step reduction according to the type of cusp in that area.



Proximal surface preparation

Using fine tapered fissure bur, the contact area should be opened carefully with the bur rested on the prepared tooth (to prevent any damage to the adjacent tooth) otherwise the rough surface lead to future caries.

Angles formed between surfaces should be rounded because this will act as a stress concentration area.



Proximal surface preparation

moving the bur up & down, the contact will be opened bucco-lingually. Once the contact is opened, a round-end tapered fissure bur is used to plane the wall while forming a chamfer finishing line.

Safe-sided disc can also be used during the proximal reduction in order to prevent any damage to the adjacent tooth. Placing a matrix band on the adjacent tooth can also be helpful.



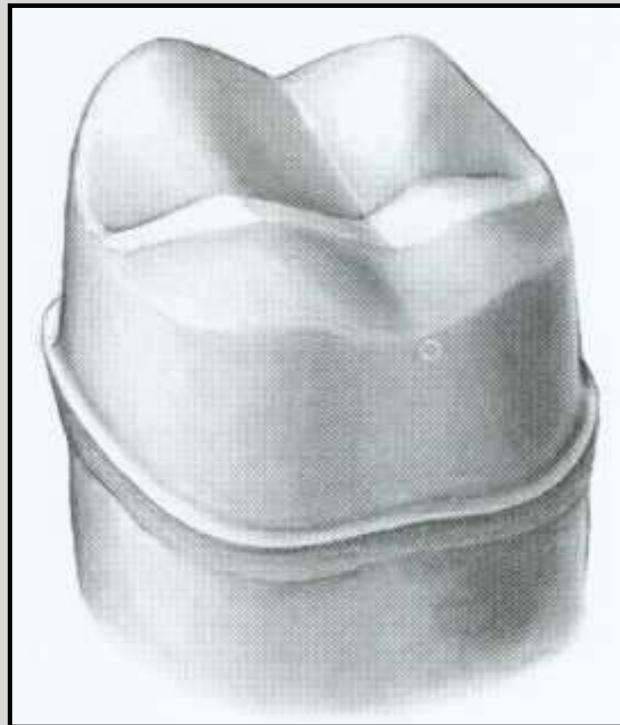
Safe-sided disc



Finishing the preparation:

After completing the preparation of the occlusal and axial surfaces sharp line-angle should be removed because this will act as a stress concentration area. Finish all prepared surfaces and slightly round all line angles.

Smoothing angles between occlusal and axial surfaces facilitates impression making, waxing, investing, and casting because bubble formation is reduced.



Use a fine-grit diamond bur of slightly greater diameter for:

1-finishing the chamfer margin.

2- The wider diamond is recommended because it will smooth out any unwanted ripples (roughness) that may be created

during axial reducing and will eliminate any unsupported enamel at the margin.



A seating groove (retentive) is placed in the buccal surface of the lower molar and the palatal surface of the upper molar

The advantages of placing a seating groove are:

1. It acts as a guide for the placement of the crown.
2. It prevents the rotation of the crown (by increasing the resistance).
3. It improves the retention.



THANK YOU

Any question





Your comment

